

Minimum Requirements Document

Axiom Optics HEGOA Table Top Motion Hexapod

3 April 2026

Scope: The AFRL/RWTSH requires the purchase of the Axiom Optics HEGOA Table Top Motion Hexapod including SYM_Motion API, UDP Acquisition + State Feedback, and External Real-Time Trajectory features.

Background: The Kinetic Kill Hardware-in-the-Loop Simulation (KHILS) Laboratory seeks to procure a bench-top hexapod motion platform to control position and angular orientation of seeker sensor prototypes for real-time hardware-in-the-loop (HWIL) simulations. This is an urgent requirement needed for integration into a HWIL testbed under development to support the Hypersonic Infrared Target Sensing (HITS) ARAP program. The hexapod will serve as a flight motion platform used to articulate a concept seeker sensor for physically simulating representative missile trajectories and line-of-sight with respect to a synthetic target. The final testbed including the hexapod represents a new capability for testing IR-guided hypersonic weapons and will be demonstrated to HITS program stakeholders and OUSW customer upon sensor integration and initial testing. Tech transition to tri-service (plus MDA) DoW test facilities would follow.

Requirements/Salient Characteristics: The following Salient Characteristics are firm requirements and any “or equal” item must meet these characteristics:

- Hexapod, controller & set of cables, SYM_Motion control software, documentation, factory acceptance tests, packaging & hotline during starting process
- Travel range Tx-Ty / Tz (mm): $\pm 100 / \pm 50$
- Travel range Rx-Ry / Rz ($^{\circ}$): $\pm 23 / \pm 30$
- Speed Tx, Ty & Tz (mm/s): ± 200
- Speed Rx, Ry & Rz ($^{\circ}$ /s): ± 50
- Acceleration Tx, Ty & Tz (mm/s²): ± 800
- Acceleration Rx, Ry & Rz ($^{\circ}$ /s²): ± 200
- Payload capacity (vertical only, kg): 50
- Position Accuracy ($^{\circ}$): 0.005
- Resolution ($^{\circ}$): 0.001
- Repeatability ($^{\circ}$): 0.005
- SYM_Motion API: controlling the hexapod through a software interface other than SYM_Motion software
- UDP Acquisition + State Feedback: broadcasting at 100 Hz of the hexapod position retrieved from motors absolute encoders information

- ERTT Feature (External Real-Time Trajectory): allows the user to control the hexapod on-the-fly by means of its own real-time trajectory generator

The Axiom Optics HEGOA hexapod system is being purchased for KHILS Laboratory research and development of new EO/IR sensing capabilities. The hexapod is required to articulate a sensor-under-test in position and angle with respect to a high-fidelity scene projection system. The hexapod will control the geometry between the sensor and target in a real-time hardware-in-the-loop simulation.